LUFA - Logging Ui for Automations

Schnelle Fehleranalyse und Compliance-Überblick für Ansible/ AWX

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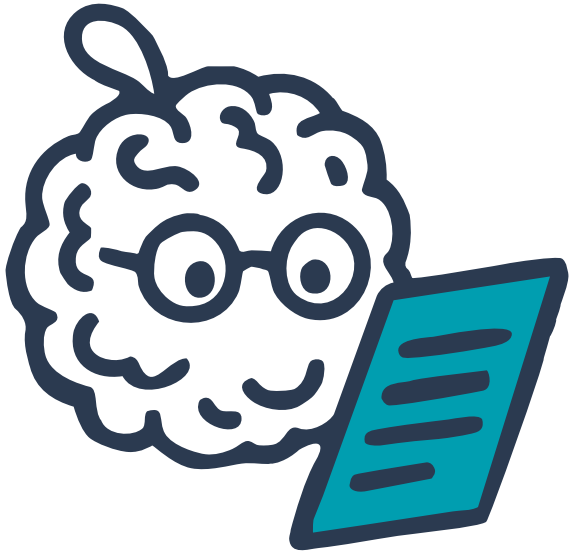

LUFA Logging-Ui For Automations

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- AWX im Rechenzentrumsbetrieb
- Probleme aus der Praxis
- Die Idee und Geschichte von LUFA
- Features
- Technologiestack & Architektur
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- Demo
- Wie geht es weiter?

Intro

- DevOps Engineer bei GISA GmbH

- IT-Komplettdienstleister
- Standorte: Halle, Leipzig, Chemnitz, Berlin, Cottbus
- BSI-zertifizierter Rechenzentrumsverbund

AWX im Rechenzentrumsbetrieb

🖥️ Serverlandschaften

- Tausende Windows-Server
- Tausende Linux/Unix-Systeme
- Hunderte Hardware-Server

🗄️ Datenbanken & SAP

- Hunderte Datenbank-Instanzen
- Hunderte SAP-Systeme

☸️ Plattformen & Applikationen

- Zahlreiche Kubernetes-Cluster
- Applikationsspezifische Automatisierungen

🌐 Netzwerk

- Erste Netzwerkgeräte

```
INFO     Processing JSON output...
INFO     Loaded XXX groups, YYYY hosts
INFO     Inventory import completed for Alle Systeme mit aktiver IP in 91.0s
```

Jobs

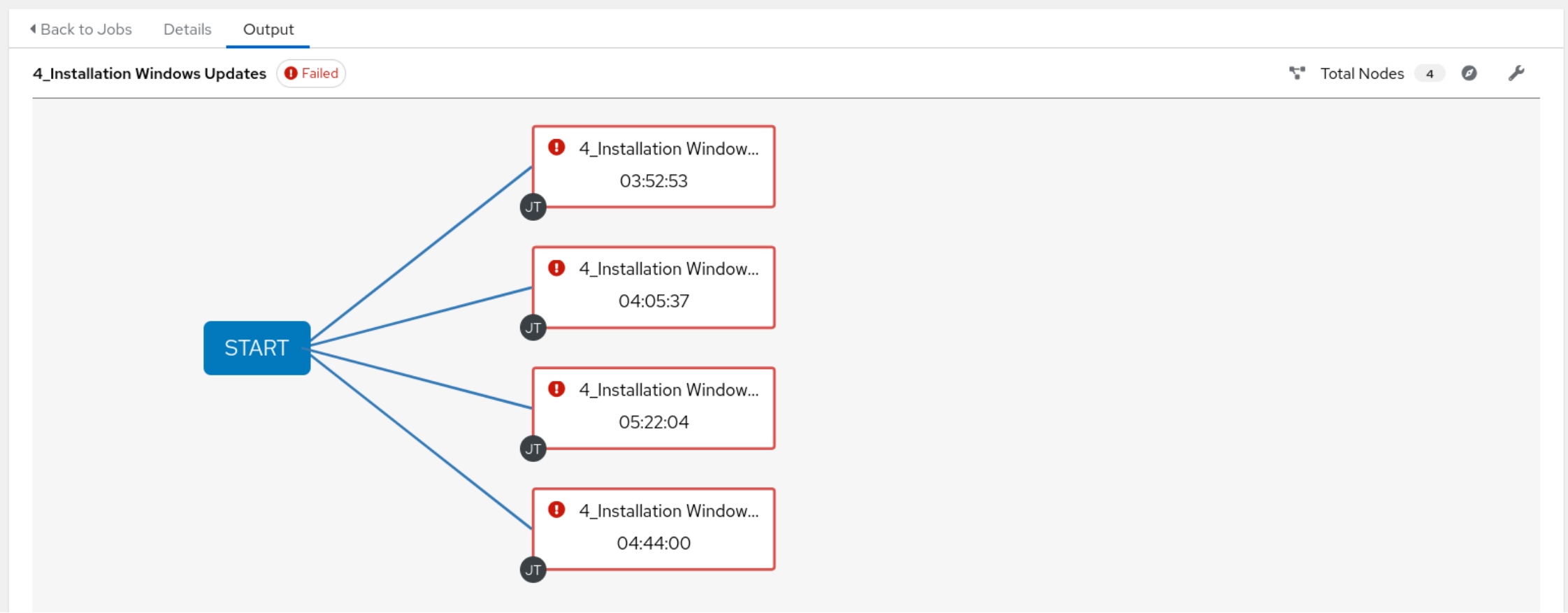
Name | Delete | Cancel jobs | 1 - 20 of 62

Name (name__icontains) zabbix × | Clear all filters

Name	Status	Type	Start Time	Finish Time	Actions
3100440 – Zabbix ausrollen	Failed	Playbook Run	3/23/2026, 4:03:34 AM	3/23/2026, 5:37:42 AM	
3098260 – Zabbix ausrollen	Failed	Playbook Run	3/22/2026, 4:03:15 AM	3/22/2026, 5:43:01 AM	
3096096 – Zabbix ausrollen	Failed	Playbook Run	3/21/2026, 4:03:21 AM	3/21/2026, 5:31:49 AM	
3093819 – Zabbix ausrollen	Failed	Playbook Run	3/20/2026, 4:03:20 AM	3/20/2026, 4:48:41 AM	
3091396 – Zabbix ausrollen	Canceled	Playbook Run	3/19/2026, 4:02:38 AM	3/19/2026, 9:32:01 PM	

- Tausende Server
- Große dynamische Inventare (CMDB)
- Lange und viele Playbooks
- Parallele Ausführung
- Regelmäßige Jobs
- → Fehler sind **normal**, die Analyse ist das Problem

Probleme aus der Praxis



Jobs > 16 - Demo Playbook

Output

Back to Jobs | Details | Output

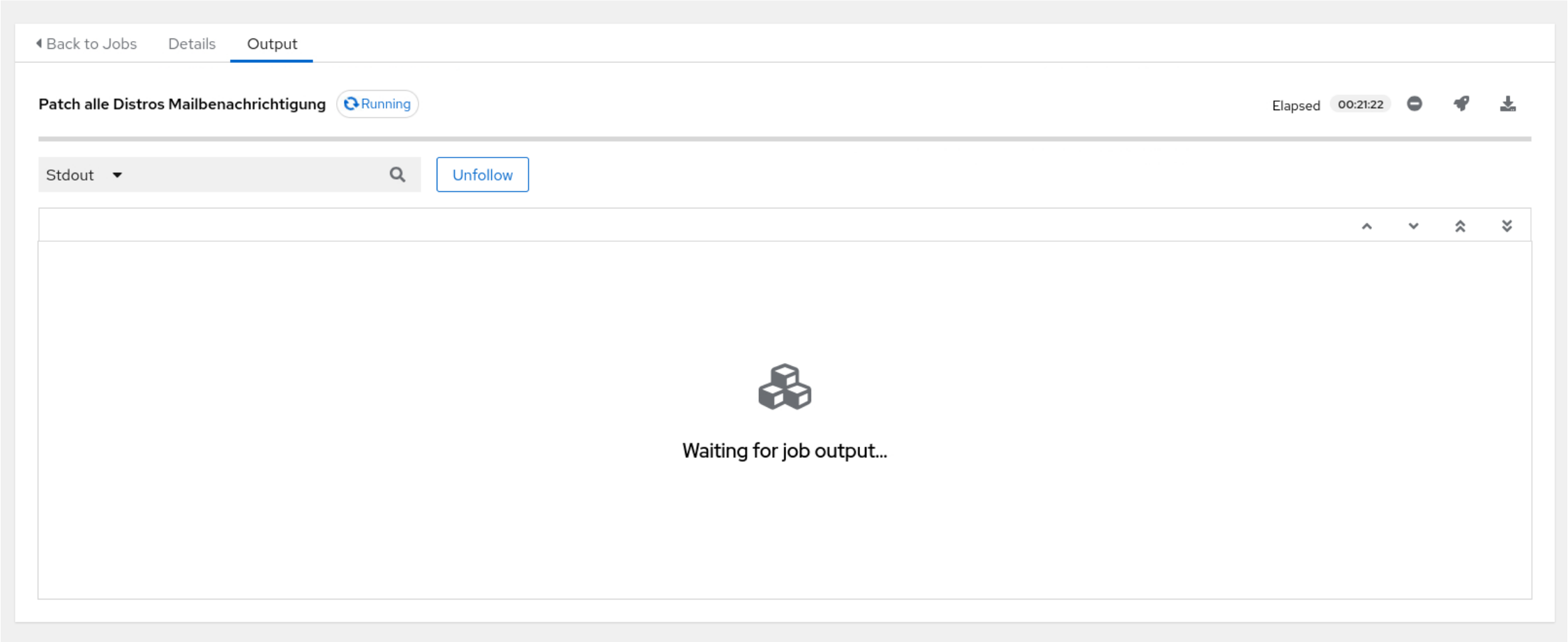
Demo Playbook Failed

Plays 2 | Tasks 67 | Hosts 9 | Failed 2 | Elapsed 00:01:51

Stdout

```
87  ok: [server2]
88
89  TASK [Tasklauf 2] ************************************************************
90  fatal: [server3]: FAILED! => {"changed": false, "msg": "ERROR", "params": {"sleep_time": 3}}
91
92  TASK [Tasklauf 3] ************************************************************
93  ok: [server5]
94  changed: [server6]
95  fatal: [server1]: FAILED! => {"changed": false, "msg": "ERROR", "params": {"sleep_time": 5}}
96
97  TASK [Tasklauf 4] ************************************************************
98  changed: [server5]
99
100 TASK [Tasklauf 2] ************************************************************
101 ok: [server8]
102 ok: [server4]
103 changed: [server7]
104
105 TASK [Tasklauf 3] ************************************************************
106 changed: [server8]
107
108 TASK [Tasklauf 4] ************************************************************
109 ok: [server6]
110 ok: [server2]
111
112 TASK [Tasklauf 5] ************************************************************
113 ok: [server5]
```

Probleme aus der Praxis

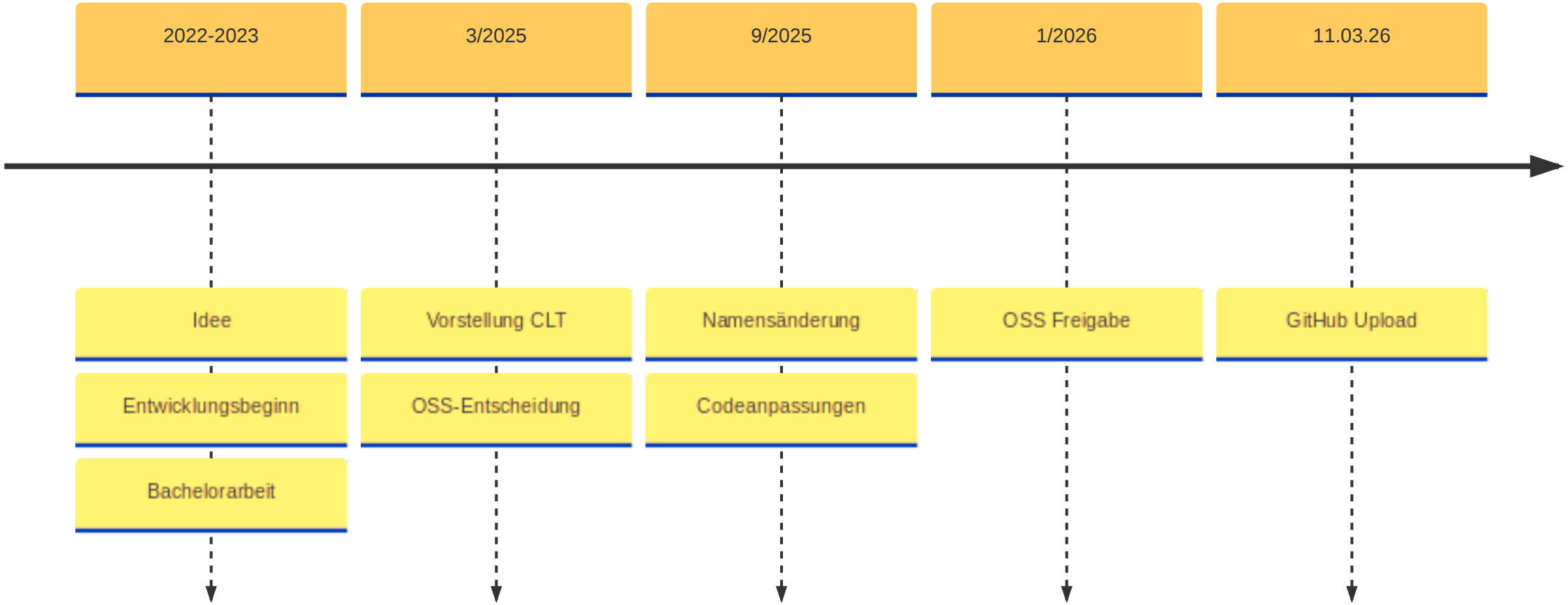


- Unübersichtlicher Output (Strategy `free` & Job Slices)
- kein Job-Output verfügbar
- hängende Tasks
- Jobs werden nur 7 Tage vorgehalten

Die Idee eines eigenen Dashboards

- schlankes Dashboard
- Fokus auf Jobs, Workflows, Hosts, Tasks
- Optimiert für die **Fehlersuche** im Ansible/AWX
- Live-Daten pro Tasks
- Überwachung von Regeljobs
- einfache Installation

Die Geschichte von LUFA



Features

LUFA Templates Workflows Jobs Hosts live ? admin

Jobs

Job ID 0 Open

Last 3 days Show

Custom Search Builder

Add Condition

Copy Excel Column visibility Share

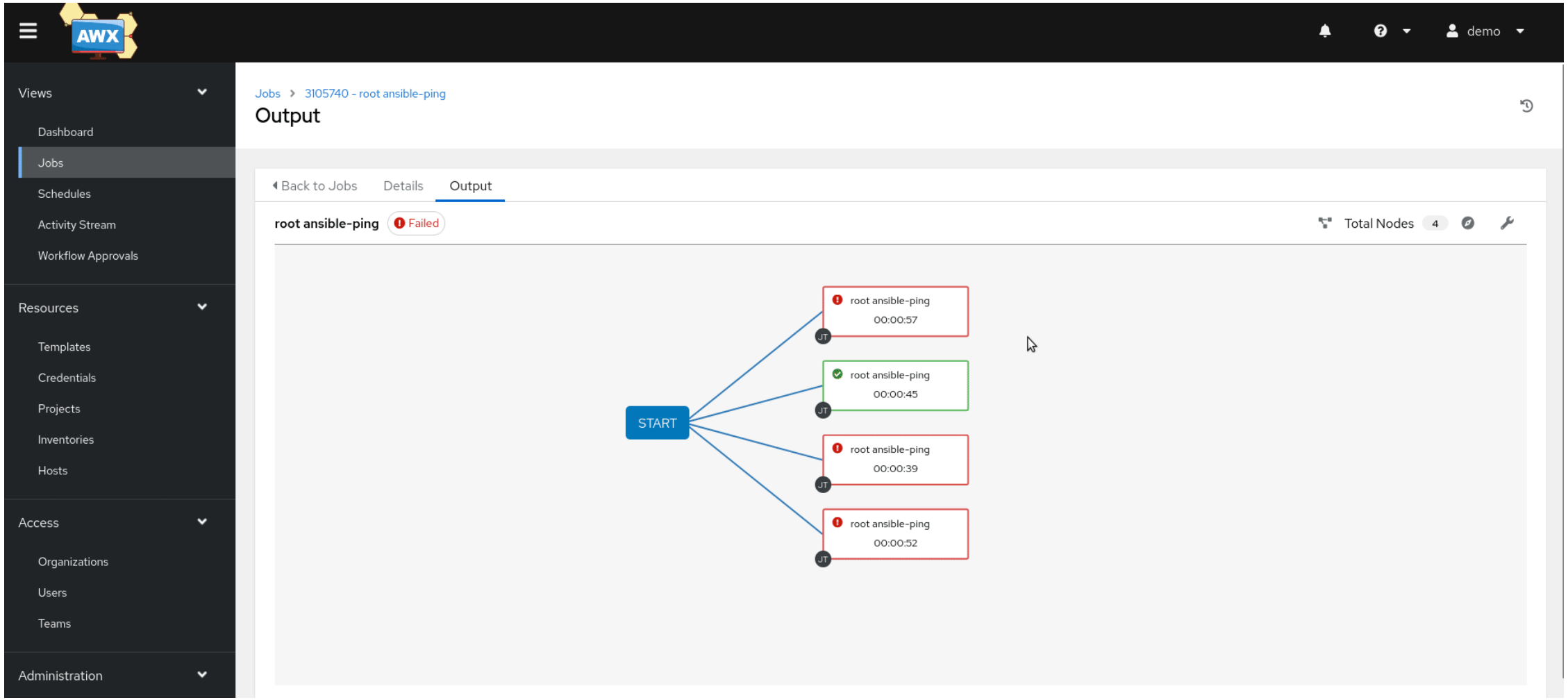
Search:

Job ID	Name	Organization	Tags	Compliance Interval in Days	User	Schedule	Workflow	Start Time	End Time	State	Limit
3103025	root ansible-ping		all	0	admin			2026-03-24 19:37:13	2026-03-24 19:37:21	failed	server1001:server1002:server1003:server1004:server...
3103010	Syslog ausrollen		all	4	admin			2026-03-24 19:27:33	2026-03-24 19:28:22	failed	server1001:server1002:server1003:server1004:server...
3102868	PW-Pol ausrollen		all	4	admin			2026-03-23 16:56:01	2026-03-23 16:56:29	failed	server1001:server1002:server1003:server1004:server...

Showing 1 to 3 of 3 entries

« ‹ 1 › »

Features


AWX
demo
Views
Dashboard
Jobs
Schedules
Activity Stream
Workflow Approvals
Resources
Templates
Credentials
Projects
Inventories
Hosts
Access
Organizations
Users
Teams
Administration
Jobs > 3105740 - root ansible-ping
Output
Back to Jobs
Details
Output
root ansible-ping
Failed
Total Nodes 4
START
root ansible-ping
00:00:57
root ansible-ping
00:00:45
root ansible-ping
00:00:39
root ansible-ping
00:00:52
JT

Compliance Feature

```
---
lufa_compliance_interval: 5
```

Ein Host gilt in einem Template als compliant, wenn dieser mindestens einmal in den gelaufenen Jobs innerhalb des Intervalls fehlerfrei durchgelaufen ist.

Compliance Feature

LUFA Templates Workflows Jobs Hosts live ? admin

Hosts

Custom Search Builder

Add Condition

Copy Excel Column visibility Limit

Search:

Host	Compliant
server1001.example.com	true
server1002.example.com	false
server1003.example.com	true
server1004.example.com	true
server1006.example.com	true
server1007.example.com	true
server1008.example.com	false
server1010.example.com	true
server1011.example.com	true
server1012.example.com	true

Showing 1 to 10 of 10 entries

« ‹ 1 › »

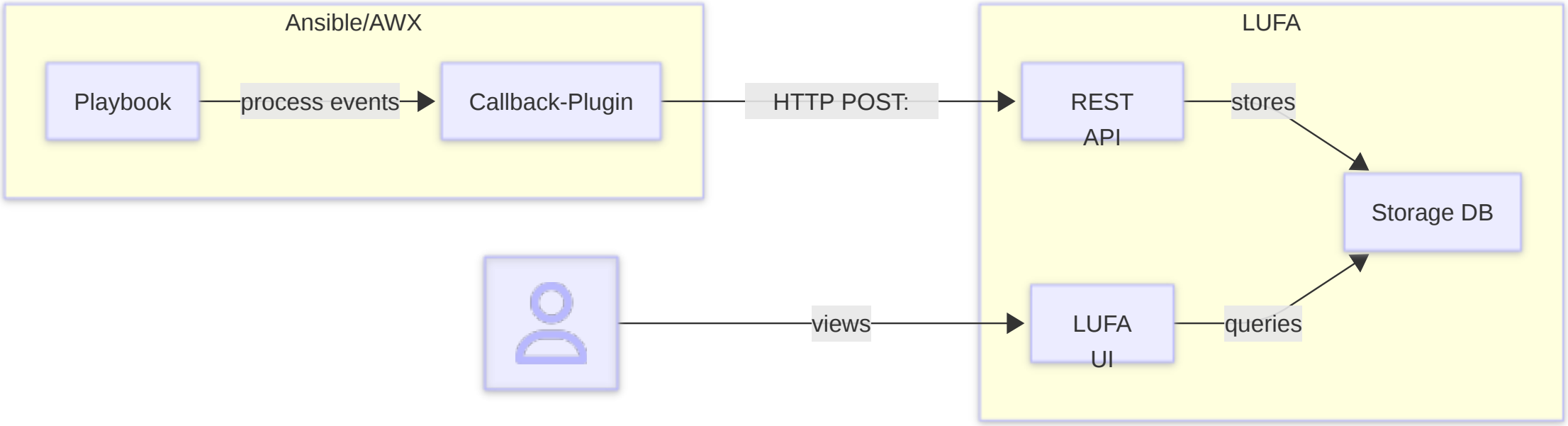
```json
{
  "server1002.example.com": [
    {
      "compliance_interval_in_days": 4,
      "last_job_run": 3103010,
      "last_successful": 0,
      "organisation": null,
      "playbook": "play-single-common-role.yml",
      "template_name": "Syslog ausrollen",
      "tower_job_template_id": 2059
    }
  ],
  "server1008.example.com": [
    {
      "compliance_interval_in_days": 4,
      "last_job_run": 3102868,
      "last_successful": 1773930792.186,
      "organisation": null,
      "playbook": "play-single-common-role.yml",
      "template_name": "PW-Pol ausrollen",
      "tower_job_template_id": 2060
    }
  ]
}
```


```
curl -H "Authorization: token <secret>" \
     -H "Content-Type: application/json" \
     https://lufa.example.com/api/v1/compliance/hosts \
| jq
```

Technologiestack

- Python
- Flask Webanwendung
- PostgreSQL oder SQLite
- LDAP oder lokale Authentifizierung
- Entwickelt für Docker & Kubernetes
- Node.js zum Laden von Abhängigkeiten

Architektur



Installation

```
# nightly
docker pull ghcr.io/gisa-oss/lufa:nightly

# released version
docker pull ghcr.io/gisa-oss/lufa:<version>
```

```yaml
services:
  lufa:
    image: ghcr.io/gisa-oss/lufa:nightly
    restart: always
    environment:
      LUFA_SECRET_KEY: "<secret>"
      LUFA_API_KEYS: "['<secret>']"
      LUFA_AWX_BASE_URL: https://awx.example.com
      LUFA_AWX_API_TOKEN: "<secret>"
      LUFA_DB_TYPE: SQLITE
      LUFA_DB_DATABASE: "/home/lufa/lufa.db"
      LUFA_LOG_FILE_PATH: /home/lufa/lufa.log
      LUFA_LOG_LEVEL: INFO
      LUFA_AUTH: LOCAL
      LUFA_AUTH_USER: admin
      LUFA_AUTH_PASSWORD: "<secret>"
    volumes:
      - "./lufa.log:/home/lufa/lufa.log:rw"
      - "./lufa.db:/home/lufa/lufa.db:rw"
    ports:
      - "8080:8080"
```

Konfiguration

```
git clone https://github.com/GISA-OSS/lufa-callback.git
cd lufa-callback

kubectl -n <awx-namespace> create configmap --from-file lufa.py lufa-callback
```

AWX: Administration > Instance Groups > default > Pod spec override

```yaml
spec:
  volumes:
  - name: callback
    configMap:
      name: lufa-callback
  containers:
  - name: worker
    volumeMounts:
      - mountPath: /usr/share/ansible/plugins/callback
        name: callback
```

Settings > Jobs > Job settings

Ansible Callback Plugins

```
[
    "/usr/share/ansible/plugins/callback"
]
```

Extra Environment Variables

```
{
  "LUFA_API_KEY": "<secret>",
  "LUFA_ENDPOINT_URIS": "https://lufa.example.com/api/v1/",
  "LUFA_REPLACE_SECRETS": "True"
}
```

Demo

https://lufa.gisa-test.de/

User: demo

Passwort: clt2026

Wie geht es weiter?

- UI-Tests
- Jobs gruppieren
- automatische DB-Migration bei Änderungen am Schema
- weitere Ansible-Ausführungsplattformen
- **Contributions Welcome!**

Vielen Dank!

- für eure Aufmerksamkeit!

- meinen Arbeitgeber
- meinen Chef
- meine Kollegen

Fragen?

🔗 Links

LUFA
https://github.com/GISA-OSS/lufa

LUFA Callback Plugin
https://github.com/GISA-OSS/lufa-callback

GISA-OSS
https://github.com/GISA-OSS

📧 Kontakt

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